



Lab 01:

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Taanis Zahra

Registration Number: FA24B1-SE-51

Course Code: BSE-665

Section: B

GitHub Username: taaniszahra51

Getting Started

 taaniszahra51	Create .gitkeep	80ddbb9 · now	 4 Commits
	Labs/Lab01/screenshots	Create .gitkeep	now
	README.md	Initial commit	now

- Demonstrates the required Labs/Lab01/screenshots/ directory structure created inside the repository.

Task 1: Create and Configure Your GitHub Account

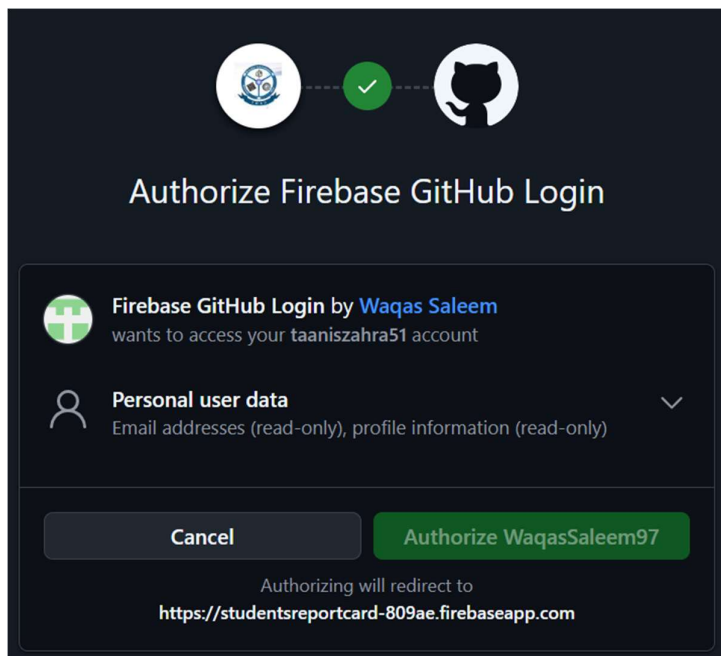
The screenshot shows the GitHub profile page for user 'taaniszahra51'. The profile is new, as indicated by the 'Joined 5 minutes ago' status. The profile picture is a green and white geometric design. The bio reads 'BS Software Engineering student at FJWU | Learning Cloud Computing & DevOps'. The 'Popular repositories' section is empty, displaying the message 'You don't have any public repositories yet.' The '1 contribution in the last year' section shows a grid with a single green square in the top right corner (September). The 'contribution graph' section explains that the first square is for joining GitHub and that more contributions lead to higher contrast colors. A link to 'Read the Hello World guide' is provided.

- Demonstrates the created public GitHub profile displaying the username taaniszahra51.

The screenshot shows the 'Public profile' settings page. The 'Name' field is set to 'Taanis Zahra'. The 'Public email' field is set to 'fa24b1-se-051@fjwu.edu.pk'. The 'Bio' field is set to 'BS Software Engineering student at FJWU | Learning Cloud Computing & DevOps'. The 'Pronouns' field is set to 'she/her'. The 'Profile picture' section shows the same green and white geometric design as the profile picture in the previous screenshot, with an 'Edit' button next to it.

- Demonstrates the completed Name field under GitHub Public Profile settings displaying the student's full actual name

Task 2: Register in the Student Marks Portal



- Demonstrates the Student Marks Portal login page prior to authenticating via GitHub.



Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

Section

Submit enrollment

Sign out

- Demonstrates student GitHub profile information and registration details integrated inside the portal.

Course enrollments

[Sign out](#)**Taanis Zahra**

2024-BSE-51

[+ Enroll in another course](#)

ENROLLMENT STATUS

Your courses

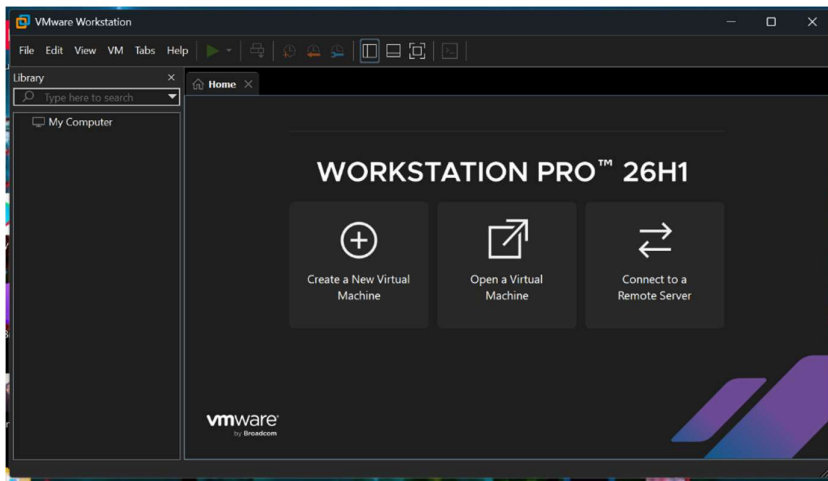
Cloud Computing (SE-665) Section B

Waiting for administrator approval

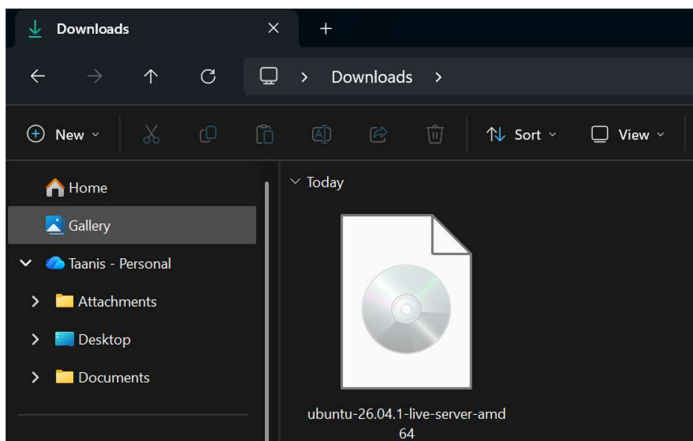
Pending approval

- Demonstrates successful submission of course enrollment awaiting approval inside the Student Marks Portal.

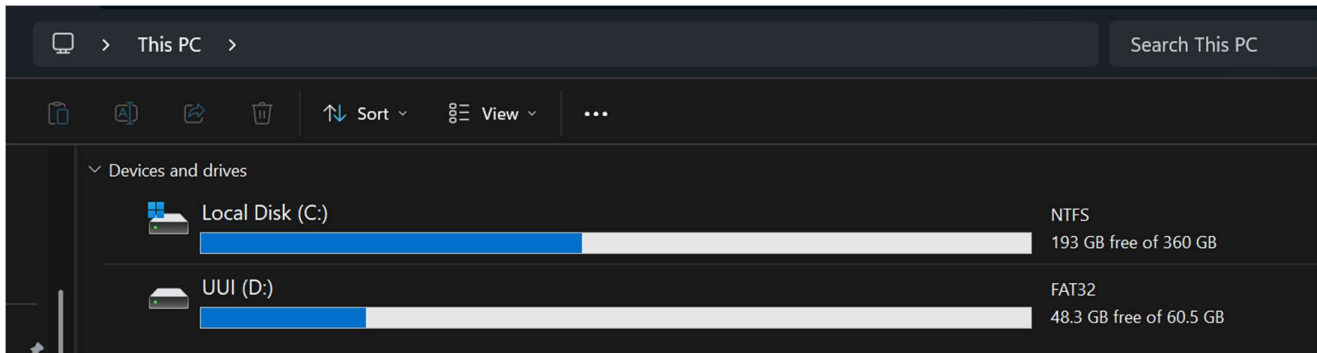
Task 3: Download VMware Workstation Pro and Ubuntu Server



- Demonstrates VMware Workstation Pro installed and running on the Windows host machine.

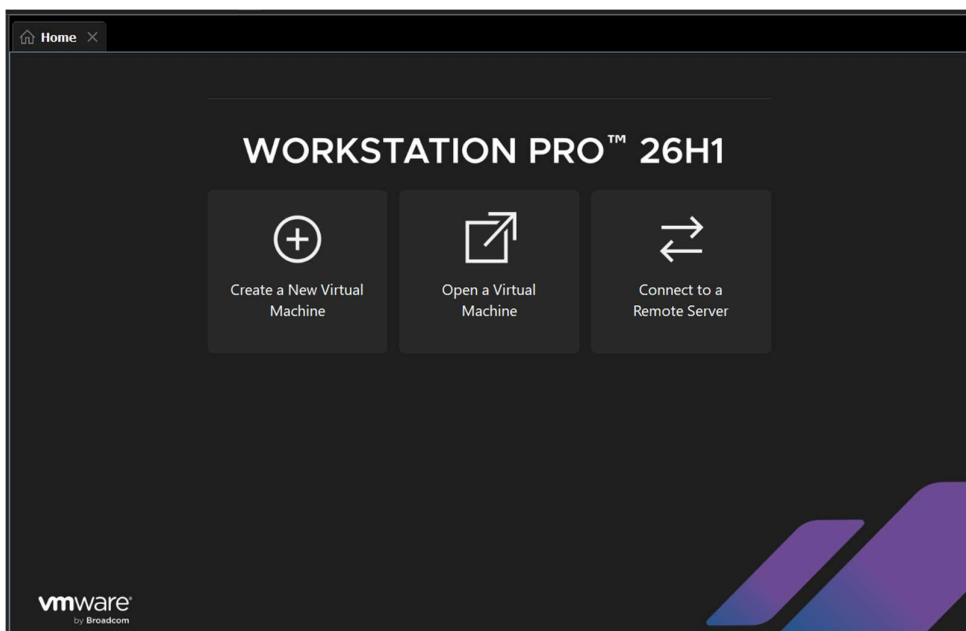


- Displays the downloaded Ubuntu Server LTS ISO file inside the host system file manager.



- Demonstrates sufficient available local disk storage (minimum 20 GB) for virtual machine creation.

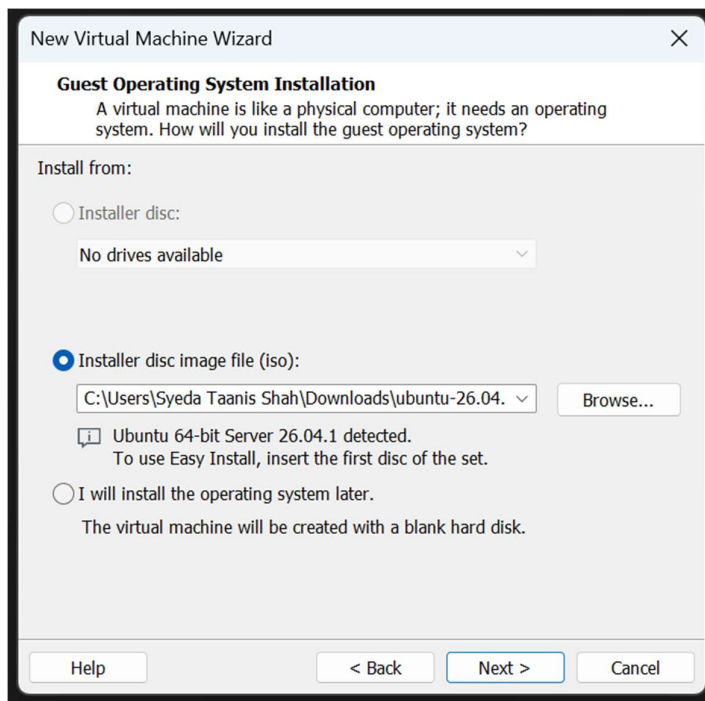
Task 4: Create the Ubuntu Server Virtual Machine



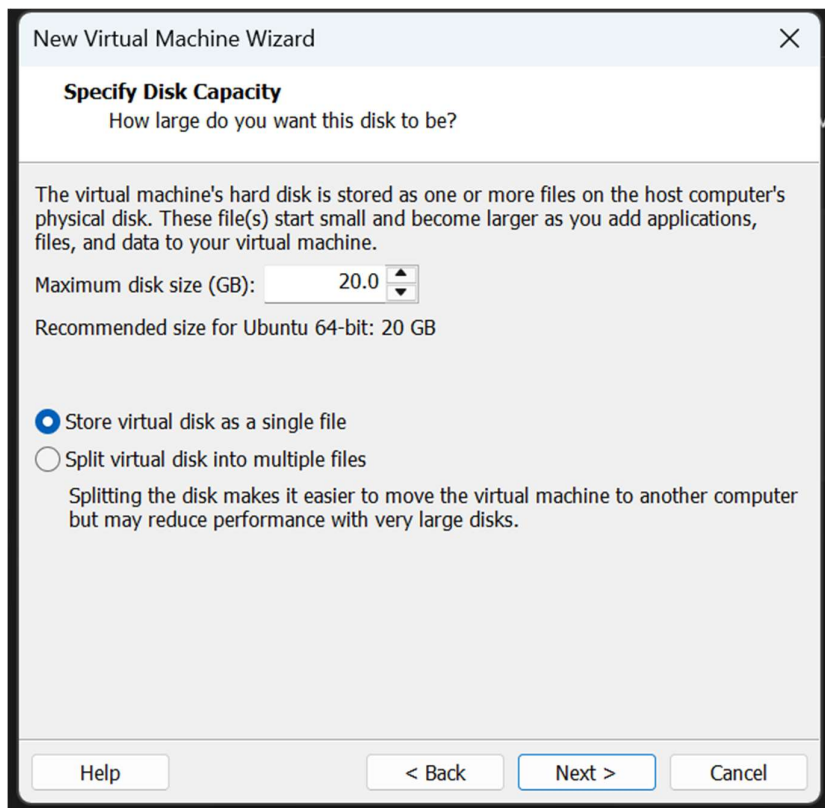
- Demonstrates opening the VMware New Virtual Machine Wizard to begin virtual machine setup.



- Demonstrates selecting the Typical configuration option for the virtual machine setup.



- Displays the selected local file path for the Ubuntu Server ISO installer image.



- Demonstrates the final virtual machine configuration summary prior to creation.

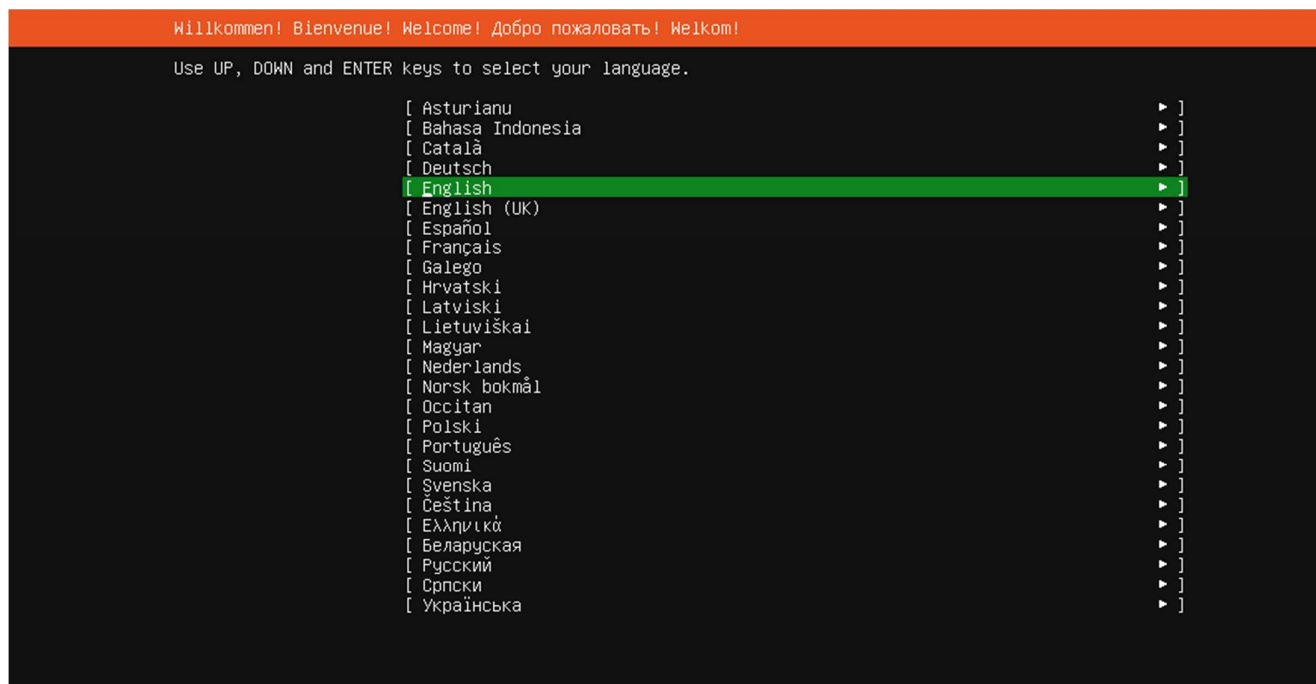
```

Begin: Configuring fstab... .. done.
Begin: Setting up locales... .. done.
Begin: Setting up automatic login... .. done.
Begin: Setting hostname... .. done.
Begin: Setting up console keyboard... .. done.
Begin: Applying desktop settings... .. done.
Begin: Regenerating SSL certificate... .. done.
Begin: Loading preseed file... .. done.
Begin: Adding live session user... .. ^[Spasswd: password changed.
done.
Begin: Setting up init... .. done.
Begin: Configuring accessibility options... .. done.
Begin: Disabling update-notifier... .. done.
Begin: Enabling detection of crashes... .. done.
Begin: Disabling unnecessary KDE services... .. done.
Begin: Disabling the lock screen for LXQt (LP: #2000787)... .. done.
Begin: Generating APT cache for cdrom ... Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 file:/cdrom resolute/main amd64 Packages [39.1 kB]
Get:5 file:/cdrom resolute/restricted amd64 Packages [13.9 kB]
^[

```

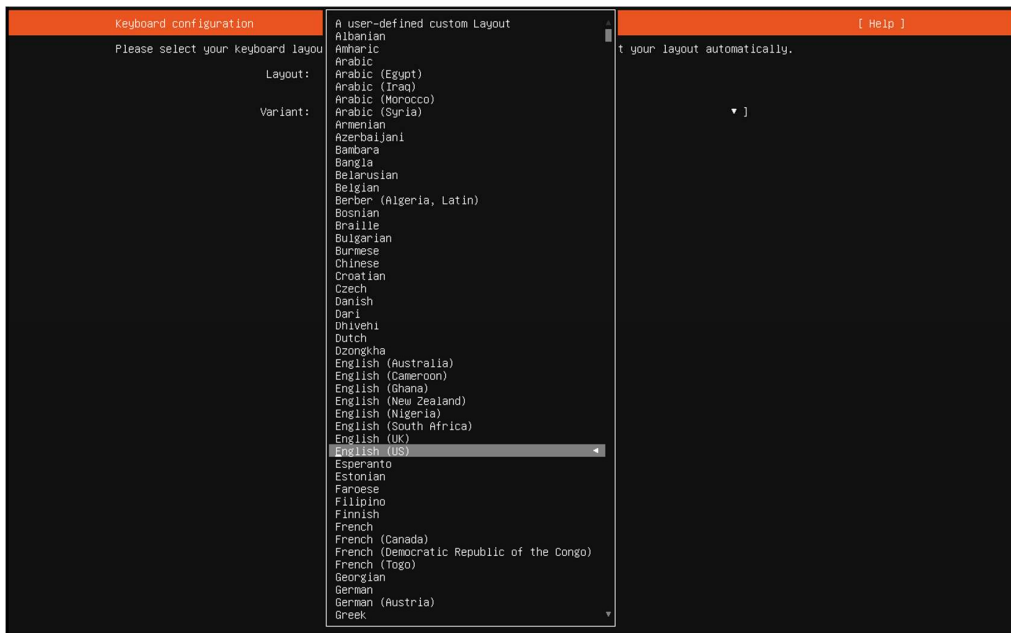
- Demonstrates the Ubuntu Server installer boot screen upon powering on the virtual machine.

Task 5: Choose the Installation Language

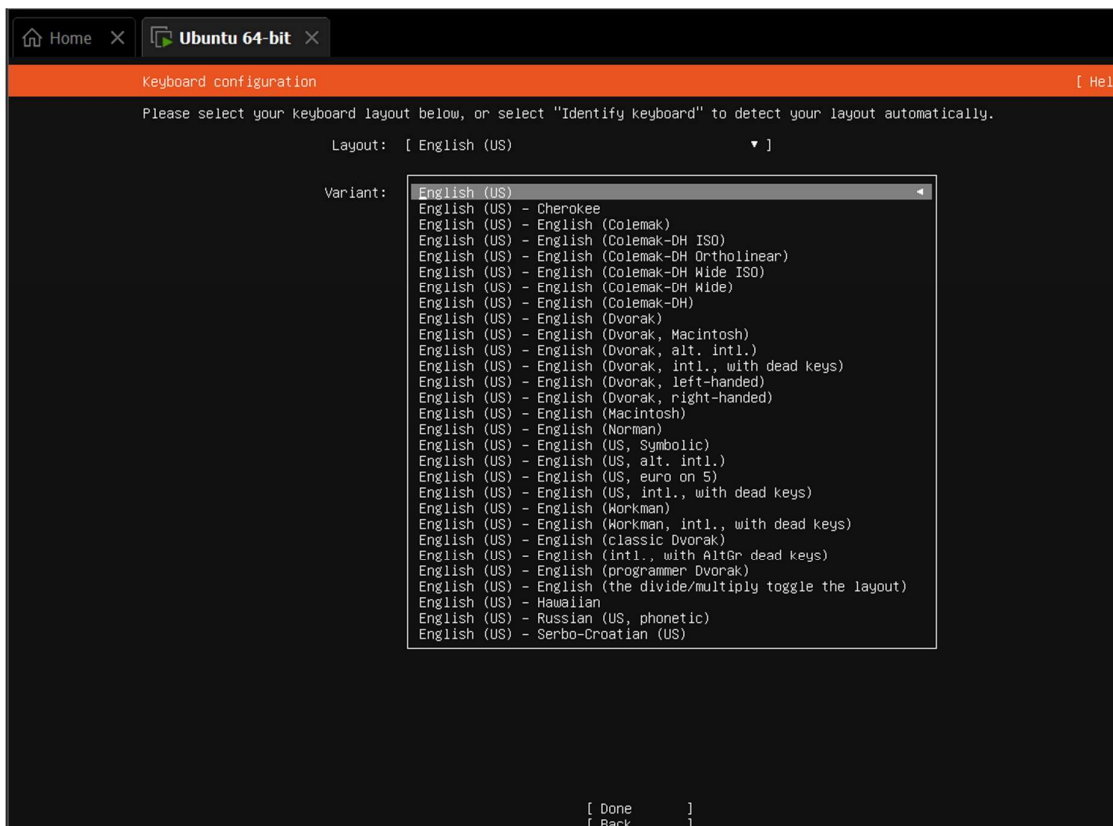


- Demonstrates selecting English as the primary installer language.

Task 6: Configure the Keyboard Layout

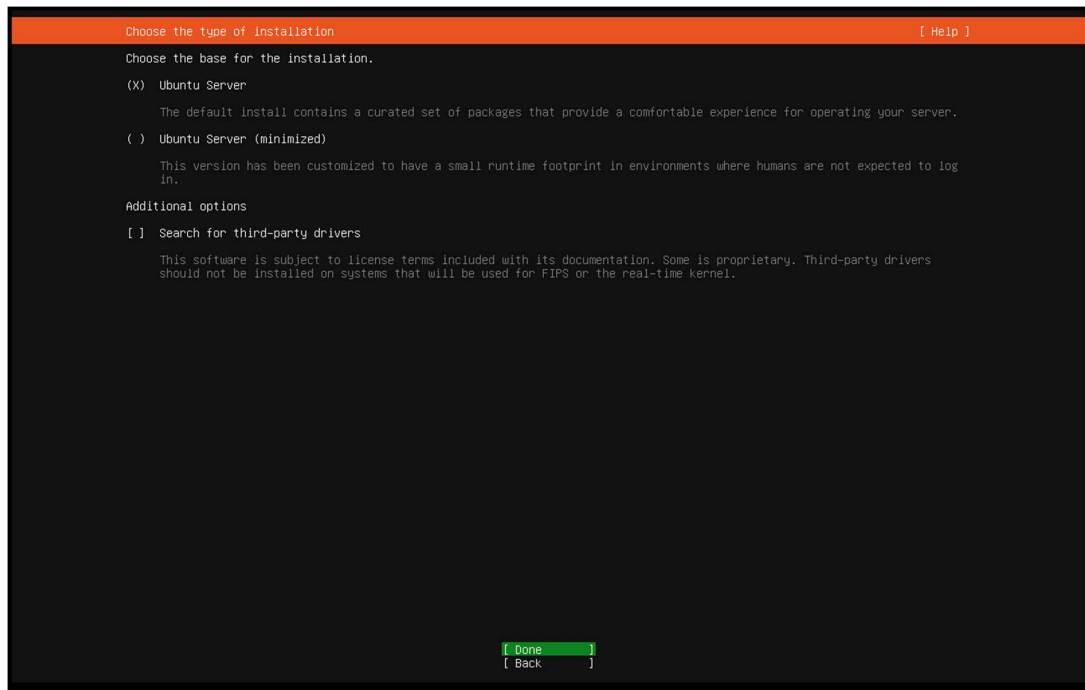


- Demonstrates selecting the primary keyboard layout (English US) during the Ubuntu Server installation setup.



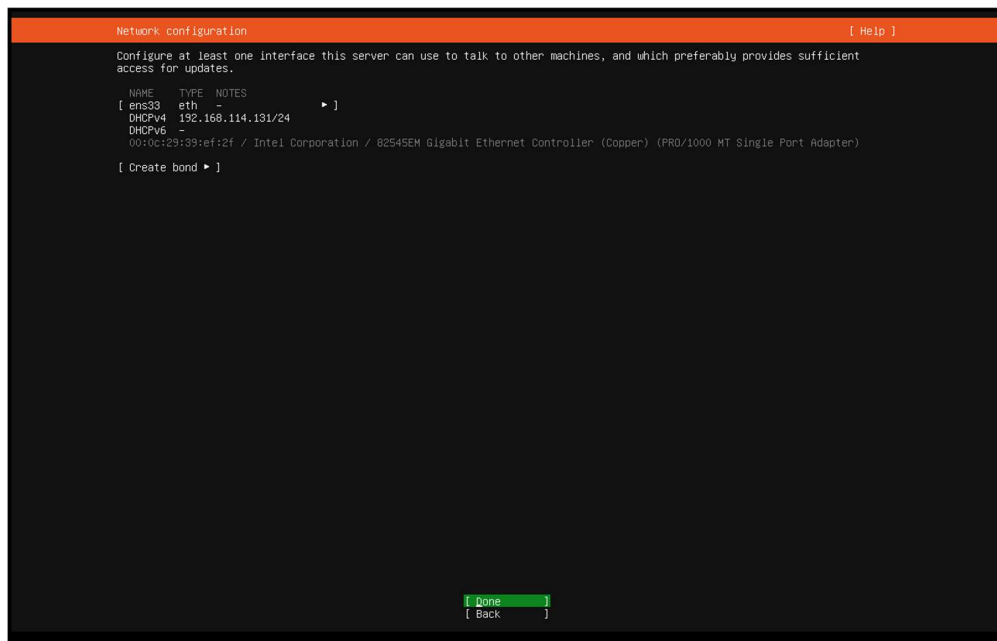
- Displays selecting the matching keyboard layout variant (English US) before proceeding with system setup.

Task 7: Choose the Ubuntu Server Installation Type

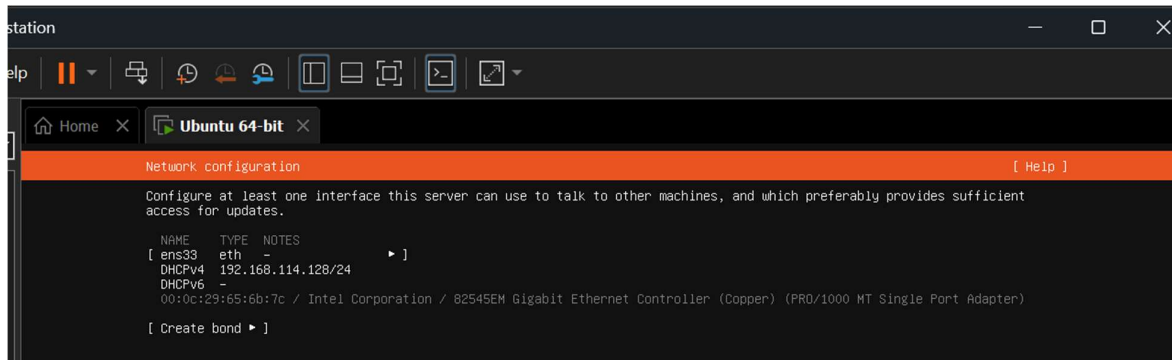


- Demonstrates selecting the standard base Ubuntu Server installation option instead of MAAS deployment.

Task 8: Configure the Network

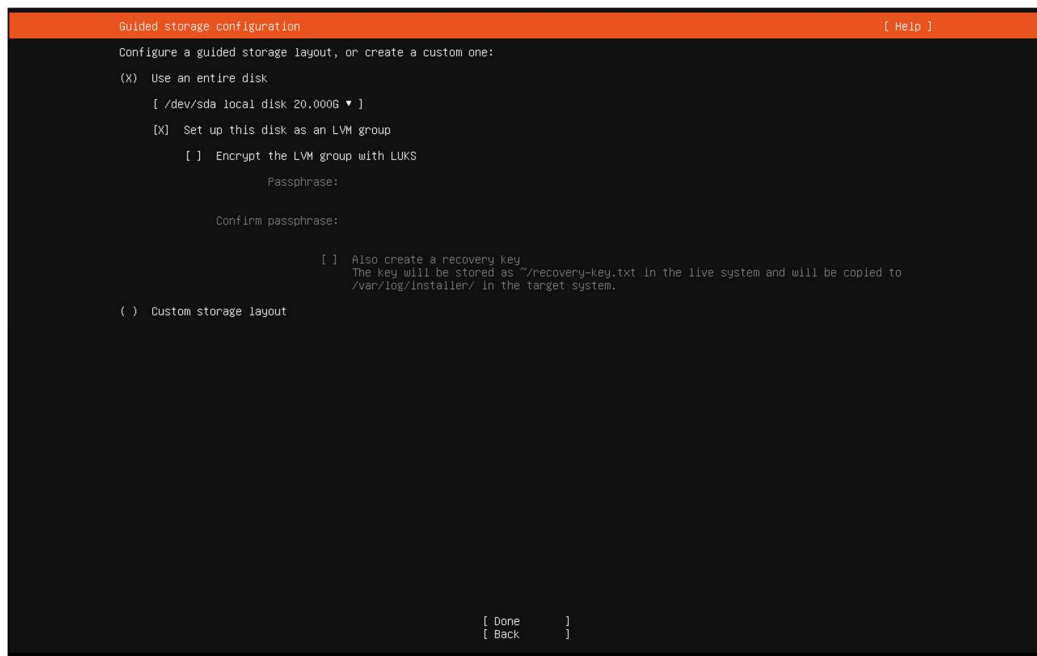


- Demonstrates detection of the primary Ethernet network interface (ens33) by the Ubuntu Server installer.

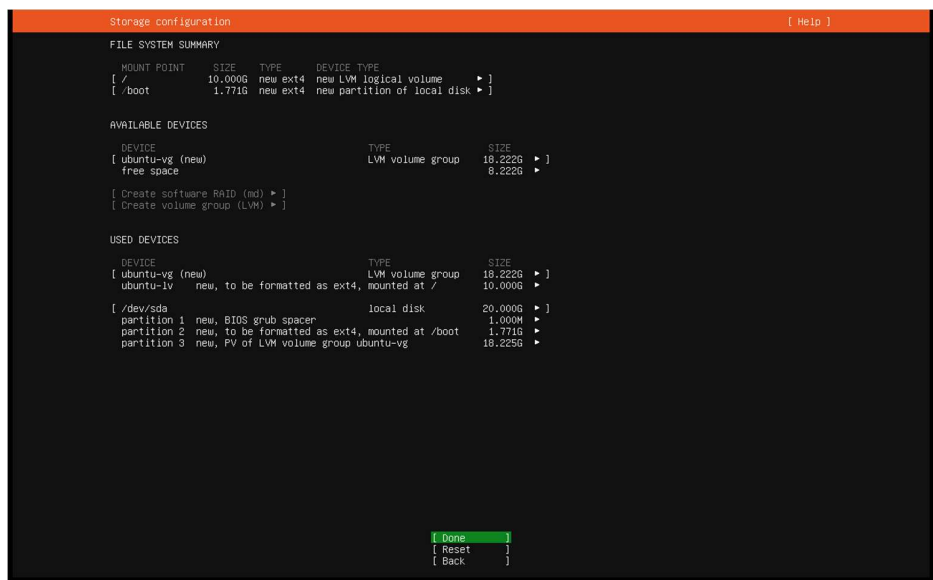


- Displays the active IPv4 address automatically assigned to the network interface via DHCP.

Task 9: Configure Storage

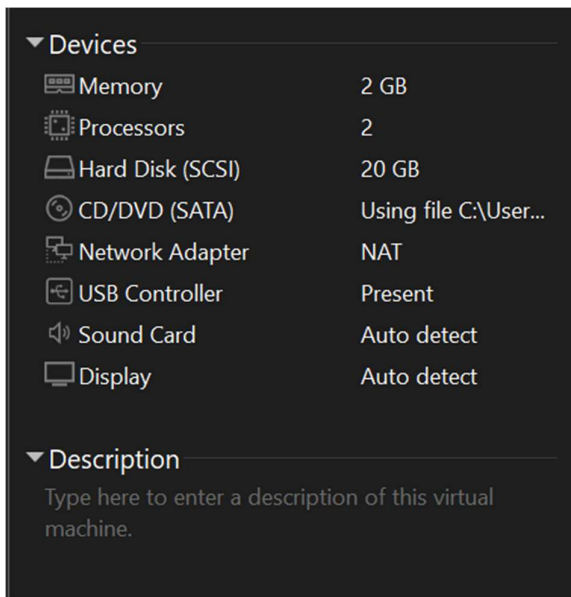


- Demonstrates selecting the guided storage option to use the allocated VMware virtual disk for installation.

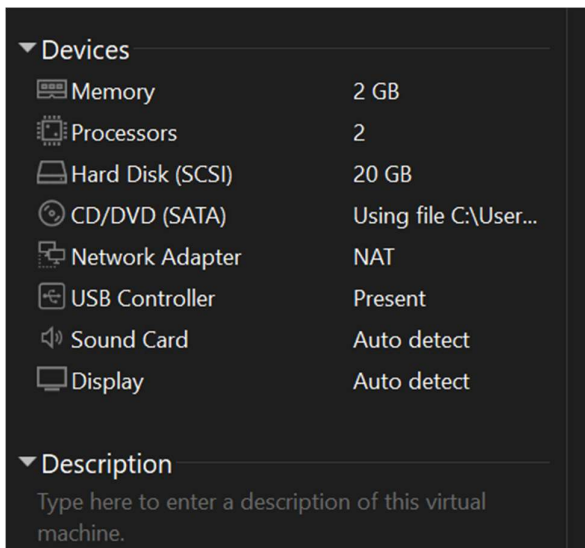


- Displays the reviewed filesystem storage layout and partition summary prior to applying changes.

Task 10: Select the Installation Disk



- Demonstrates selecting the primary target virtual disk for the Ubuntu Server installation.



- Displays the target virtual disk name and allocated capacity to verify installation on the correct VMware storage.

Task 11: Review the Partitions

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

Maximum recommended memory
(Memory swapping may occur beyond this size.)
5.9 GB

Recommended memory
2 GB

Guest OS recommended minimum
1 GB

- Demonstrates the proposed partition layout and filesystem summary on the target VMware virtual disk.

Task 12: Confirm the Storage Changes

Filesystem setup

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
/	19.997G	ext4	partition of local disk

AVAILABLE DEVICES

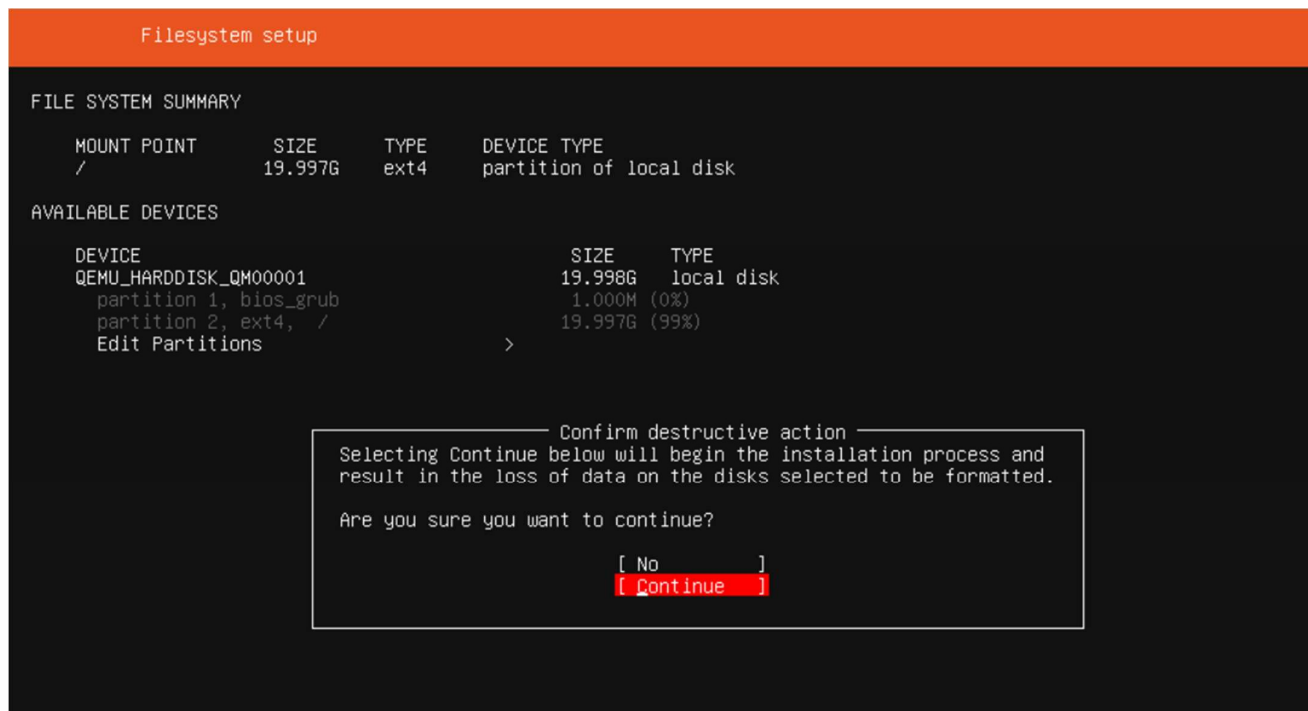
DEVICE	SIZE	TYPE
QEMU_HARDDISK_QM000001	19.998G	local disk
partition 1, bios_grub	1.000M (0%)	
partition 2, ext4, /	19.997G (99%)	

Edit Partitions >

Confirm destructive action
Selecting Continue below will begin the installation process and result in the loss of data on the disks selected to be formatted.
Are you sure you want to continue?

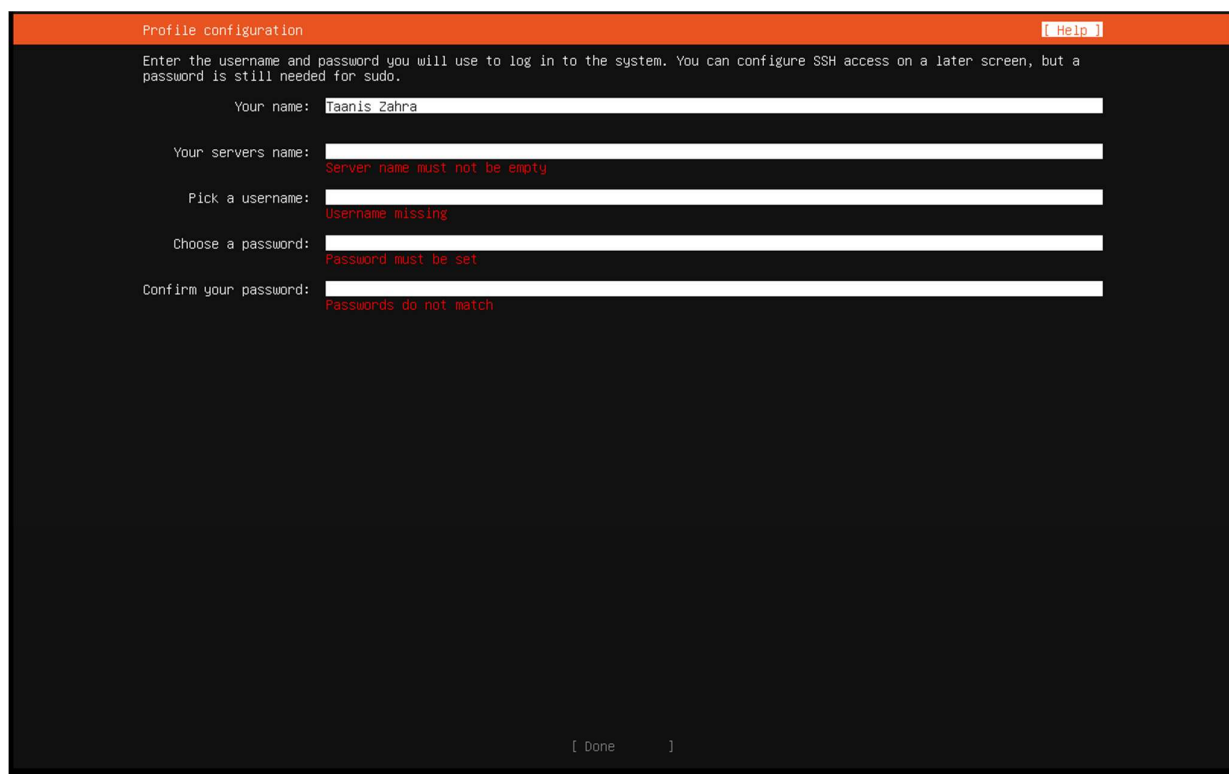
[No]
[Continue]

- Displays the installer's destructive-action warning prompt prior to overwriting the virtual disk.



- Demonstrates confirming the storage action to proceed with disk partitioning and formatting.

Task 13: Configure the Ubuntu Profile



- Demonstrates entering the student's full actual name in the profile configuration screen.

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Taanis Zahra

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: Username missing

Choose a password: Password must be set

Confirm your password: Passwords do not match

[Done]

- Demonstrates setting the server hostname to ubuntu during system configuration.

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Taanis Zahra

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: taaniszahra51

Choose a password: Password must be set

Confirm your password: Passwords do not match

[Done]

- Demonstrates configuring the primary system username to match the student's GitHub handle (taaniszahra51).

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

- Demonstrates the fully completed profile setup page featuring hostname and user credentials.

Task 14: Complete the Software Installation

Updating system

[Help]

```

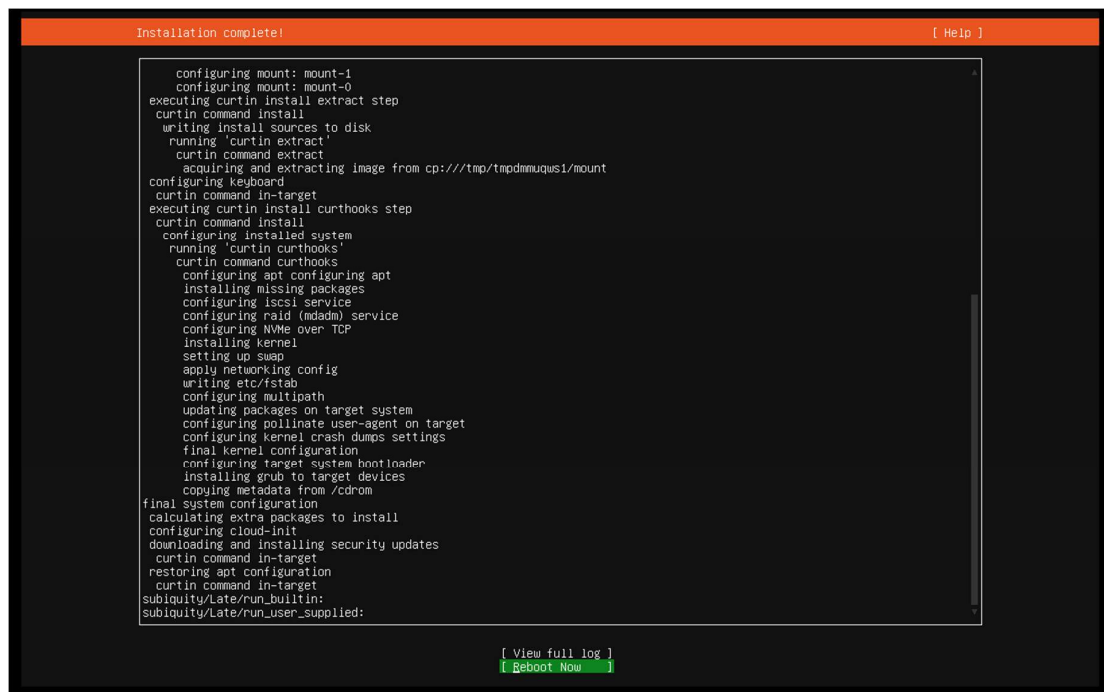
configuring partition: partition-2
configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpdmuqes1/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multiboot
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target

```

[View full log]

- Displays the ongoing system installation progress and package extraction on the virtual machine.

Task 15: Reboot and Verify the Ubuntu Identity

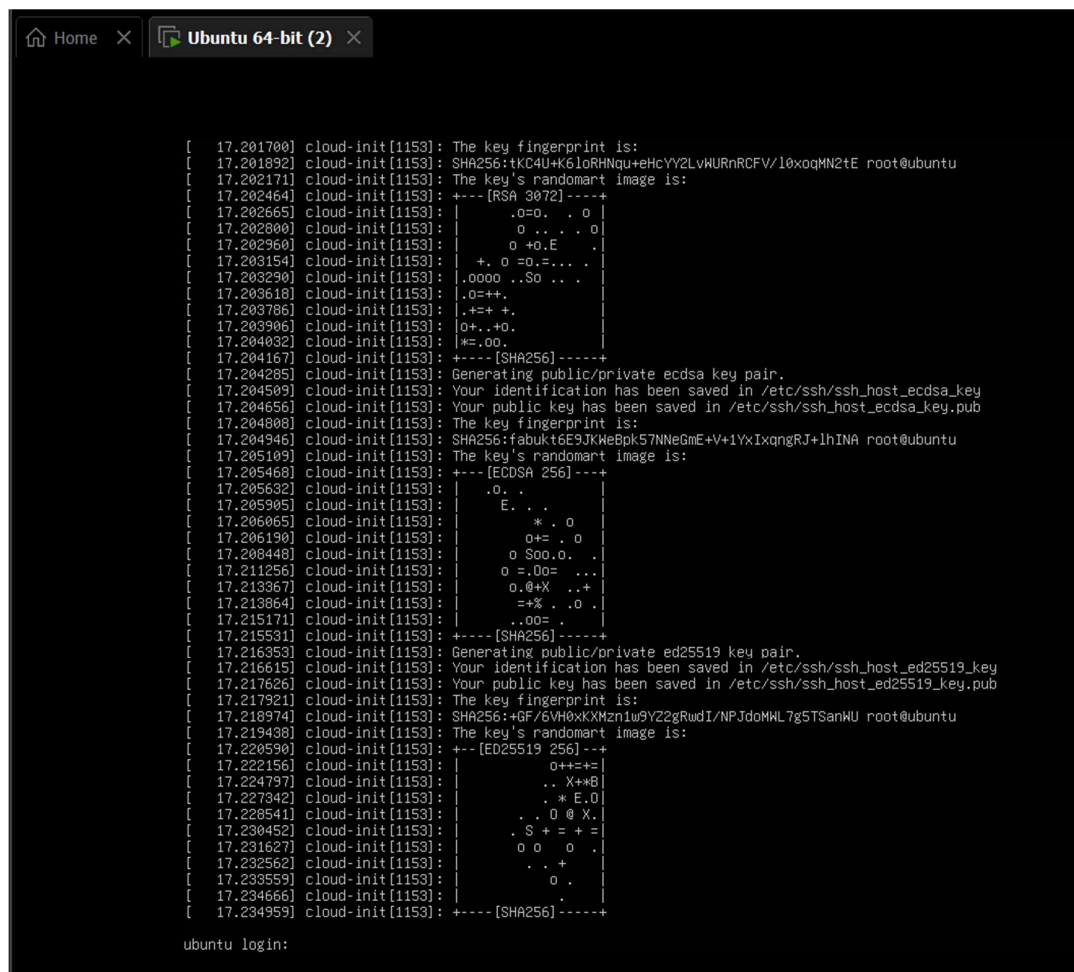


```
Installation complete! [ Help ]

configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpdmuqsl/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloder
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_built-in:
subiquity/late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

- Demonstrates the installation complete notification prior to system reboot.



```
Home x Ubuntu 64-bit (2) x

[ 17.201700] cloud-init[1153]: The key fingerprint is:
[ 17.201892] cloud-init[1153]: SHA256:tkD4U+K6loRHNqu+eHcYY2LvMURnRCFV/10x0qMN2tE root@ubuntu
[ 17.202171] cloud-init[1153]: The key's randomart image is:
[ 17.202464] cloud-init[1153]: +---[RSA 3072]-----+
[ 17.202665] cloud-init[1153]: |.o=. . o |
[ 17.202800] cloud-init[1153]: | o .. . o |
[ 17.202960] cloud-init[1153]: | o +0.E . |
[ 17.203154] cloud-init[1153]: | +. o =0.=... |
[ 17.203290] cloud-init[1153]: |.0000 ..So .. |
[ 17.203618] cloud-init[1153]: |.o=++ |
[ 17.203786] cloud-init[1153]: |.+=+ +. |
[ 17.203906] cloud-init[1153]: |o+..+o |
[ 17.204032] cloud-init[1153]: |*=.00. |
[ 17.204167] cloud-init[1153]: +---[SHA256]-----+
[ 17.204285] cloud-init[1153]: Generating public/private ecdsa key pair.
[ 17.204509] cloud-init[1153]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 17.204656] cloud-init[1153]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 17.204808] cloud-init[1153]: The key fingerprint is:
[ 17.204946] cloud-init[1153]: SHA256:fabukt6E9JKWeBpk57NNeGmE+V+1YxIxpqRJ+IhINA root@ubuntu
[ 17.205109] cloud-init[1153]: The key's randomart image is:
[ 17.205468] cloud-init[1153]: +---[ECDSA 256]-----+
[ 17.205632] cloud-init[1153]: |.o. . |
[ 17.205905] cloud-init[1153]: | E. . |
[ 17.206065] cloud-init[1153]: | * . o |
[ 17.206190] cloud-init[1153]: | o =. . o |
[ 17.208448] cloud-init[1153]: | o S0o. . |
[ 17.211256] cloud-init[1153]: | o =.00= . |
[ 17.213367] cloud-init[1153]: | o.0+X .+ |
[ 17.213864] cloud-init[1153]: | =+X . .o |
[ 17.215171] cloud-init[1153]: | .00= |
[ 17.215531] cloud-init[1153]: +---[SHA256]-----+
[ 17.216353] cloud-init[1153]: Generating public/private ed25519 key pair.
[ 17.216615] cloud-init[1153]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 17.217626] cloud-init[1153]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 17.217921] cloud-init[1153]: The key fingerprint is:
[ 17.218974] cloud-init[1153]: SHA256:+GF/6Vh0xKXmZniw9Y22gRwdI/NPjdoMML7g5TSanWU root@ubuntu
[ 17.219438] cloud-init[1153]: The key's randomart image is:
[ 17.220590] cloud-init[1153]: +---[ED25519 256]-----+
[ 17.222156] cloud-init[1153]: | o+==+ |
[ 17.224797] cloud-init[1153]: | .. X++8 |
[ 17.227342] cloud-init[1153]: | * E.O |
[ 17.228541] cloud-init[1153]: | . . 0 @ X |
[ 17.230452] cloud-init[1153]: | . S + = + = |
[ 17.231627] cloud-init[1153]: | o o o . |
[ 17.232562] cloud-init[1153]: | . . + |
[ 17.233559] cloud-init[1153]: | o . |
[ 17.234666] cloud-init[1153]: | . |
[ 17.234959] cloud-init[1153]: +---[SHA256]-----+

ubuntu login:
```

- Displays the fresh Ubuntu Server TTY login prompt after booting up.

```

ubuntu login:
ubuntu login: taaniszahra51
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 05:43:03 PM UTC 2026

System load:  0.24           Processes:            251
Usage of /:   50.5% of 9.75GB Users logged in:      0
Memory usage: 22%           IPv4 address for ens33: 192.168.114.132
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

taaniszahra51@ubuntu:~$

```

- Demonstrates successful local terminal login using user credentials.

```

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

taaniszahra51@ubuntu:~$ _

```

- Verifies system identity displaying terminal prompt taaniszahra51@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

```

taaniszahra51@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:39:ef:2f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2939ef2f
    inet 192.168.114.132/24 metric 100 brd 192.168.114.255 scope global dynamic ens33
        valid_lft 1505sec preferred_lft 1505sec
    inet6 fe80::20c:29ff:fe39:ef2f/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
taaniszahra51@ubuntu:~$

```

- Demonstrates executing the ip addr command inside the Ubuntu Server terminal.

```

taaniszahra51@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:39:ef:2f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2939ef2f
    inet 192.168.114.132/24 metric 100 brd 192.168.114.255 scope global dynamic ens33
        valid_lft 1505sec preferred_lft 1505sec
    inet6 fe80::20c:29ff:fe39:ef2f/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever

```

- Displays the assigned network interface IPv4 address alongside the taaniszahra51@ubuntu:~\$ identity prompt.

Task 17: Connect to Ubuntu Server from Windows

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> ssh taaniszahra51@192.168.114.132
```

- Demonstrates initiating an SSH remote connection from Windows terminal to the Ubuntu Server.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> ssh taaniszahra51@192.168.114.132
The authenticity of host '192.168.114.132 (192.168.114.132)' can't be established.
ED25519 key fingerprint is SHA256:+GF/6VH0xKXMzn1w9YZ2gRwdI/NPJdoMWL7g5TSanWU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

- Displays accepting the host key fingerprint to establish a trusted SSH connection.

```
PS C:\WINDOWS\system32> ssh taaniszahra51@192.168.114.132
The authenticity of host '192.168.114.132 (192.168.114.132)' can't be established.
ED25519 key fingerprint is SHA256:+GF/6VH0xKXMzn1w9YZ2gRwdI/NPJdoMWL7g5TSanWU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.114.132' (ED25519) to the list of known hosts.
taaniszahra51@192.168.114.132's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 05:59:02 PM UTC 2026

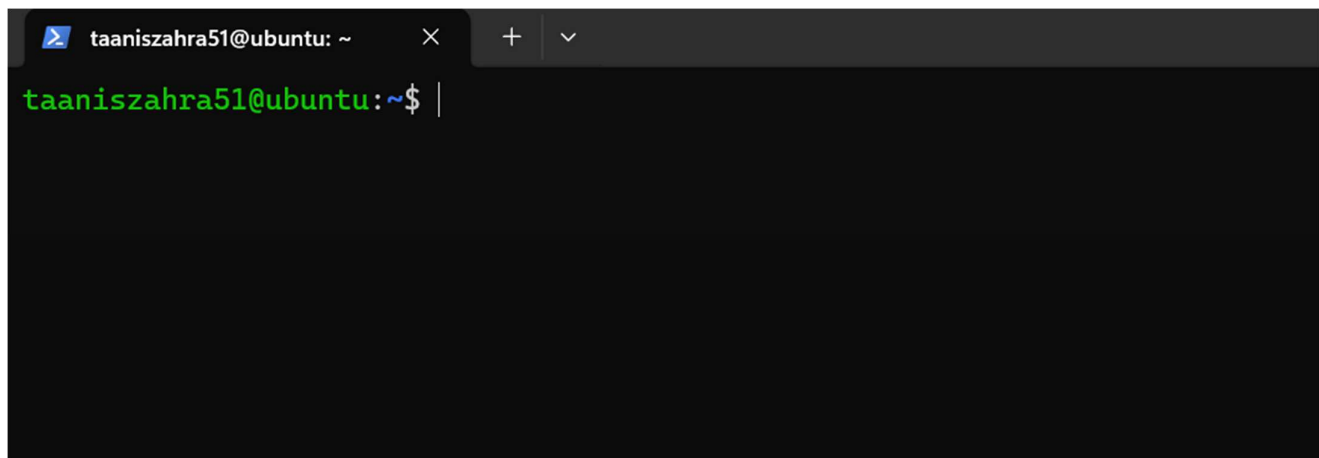
System load:  0.02               Processes:            224
Usage of /:   50.8% of 9.75GB    Users logged in:     0
Memory usage: 25%               IPv4 address for ens33: 192.168.114.132
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

- Demonstrates successful authentication and remote terminal login to Ubuntu Server over SSH.

A terminal window with a dark background. The title bar at the top shows a blue icon, the text 'taaniszahra51@ubuntu: ~', a close button 'X', and window control buttons '+', 'v'. The terminal content shows the prompt 'taaniszahra51@ubuntu:~\$' in green text, followed by a vertical cursor bar. The rest of the terminal area is empty.

```
taaniszahra51@ubuntu: ~  
taaniszahra51@ubuntu:~$ |
```

- Displays the remote shell prompt `taaniszahra51@ubuntu:~$` confirming active SSH session identity.